

MINI ASSIGNMENT

Deconstructing HCI & Computer Graphics in Daily Software

Software selected: Google Earth **Topic:** HCI vs. Computer Graphics

1. HCI FOCUS — INTERACTION ELEMENTS



Figure 1. Annotated view showing the three main HCI elements used for interaction.

HCI Analysis

The HCI part focuses on how a user controls and understands the application. In the screenshot, I identified three simple interaction elements:

Element	Purpose
1. Search box	Lets the user quickly enter a place or location instead of finding it manually.
2. Zoom control	Allows the user to move closer to or farther away from the map.
3. Navigation / rotate control	Lets the user change the viewing direction and explore the 3D scene.

2. COMPUTER GRAPHICS FOCUS — VISUAL / RENDERING ELEMENTS

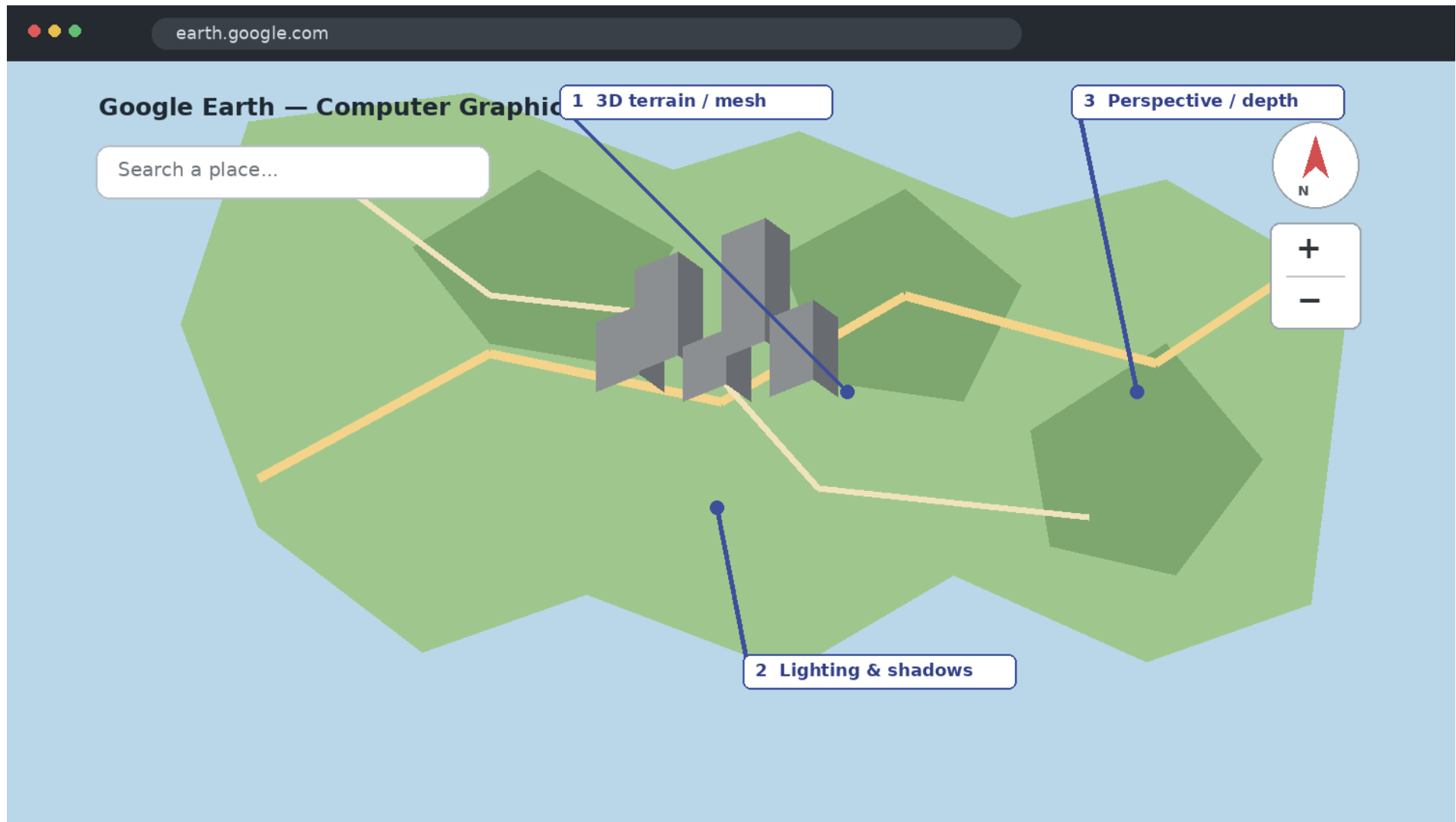


Figure 2. Annotated view showing three graphics elements that help create a 3D environment.

Computer Graphics Analysis

Element	How it creates 3D space
1. 3D terrain / mesh	Terrain and buildings are represented as three-dimensional shapes, so the environment is not flat.
2. Lighting & shadows	Changes in light and shadow help show the height, position, and shape of objects.
3. Perspective / depth	Farther objects appear smaller, which gives a stronger sense of distance and scale.

3. SHORT REFLECTION

Q1. How does Computer Graphics quality help or hinder interaction?

Good 3D graphics make it easier to understand distance, height, and the shape of places. Realistic lighting, terrain, and perspective make the map easier to understand visually. However, detailed graphics can make the application slower on a less powerful device.

Q2. What simple HCI change would I make?

I would make the navigation controls a little more noticeable and keep the most-used controls together. This would help a new user find the controls faster without changing the main design.

Conclusion. HCI and Computer Graphics have different roles but work together. HCI makes the software easier to control, while CG makes the 3D environment understandable and visually realistic.

Name: _____ Roll No: _____ Course: HCI & Computer Graphics